ZZ-H-428b
JUNE 19, 1961
SUPERSEDING
Int. Fed. Spec. ZZ-H-00428a (Army-Ord)
November 21, 1958 and
Fed. Spec. ZZ-H-428
August 30, 1948
(See 6.3.1)

FEDERAL SPECIFICATION

HOSE, RUBBER: PREFORMED AND STRAIGHT; HOSE, PREFORMED: FLEXIBLE, WIRE REINFORCED. (FOR THE COOLANT SYSTEMS OF AUTOMOTIVE AND OTHER LIQUID-COOLED ENGINES)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers preformed (curved) and straight hose; and wire-reinforced hose, for the coolant systems and heaters of automotive and other liquid-cooled engines (see 6.1).

1.2 Classification.

1.2.1 *Types and grades.* Coolant-system and heater hose covered by this specification shall be of the following types and grades as specified (see 6.1, 6.2, and 6.6):

Type I—Heavy duty, light wall.
Type II—Heavy duty, heavy wall.
Type III—Moderate duty, light wall.
Type IV—Moderate duty, heavy wall.
Type V—Normal service.
Type VI—Heater hose.
Type VII—Wire reinforced.

Grade A—Suitable for extreme low temperature service down to minus 65° F., and fungus resistant.

Grade B—Suitable for low temperature service down to minus 40° F.

1.2.2 *Material.* Material used in the hose shall be of the following classes as specified (see 3.2, 6.2, 6.3, 6.3.1, and table I).

Class 1—Oil resistant, low volume swell.
Class 2—Oil resistant, medium volume swell, and ozone resistant.
Class 3—Nonoil resistant, high quality.
Class 4—High temperature resistant, nonoil resistant, and ozone resistant.

1.2.2.1 *Applicability of classes to types and grades.* On hose covered by this specification, classes shall be applicable to types and grades as shown in table I.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

CCC-T-191—Textile Test Methods.

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TABLE I.—*Applicability of classes to types and grades*

Types	Grades	Classes
I	A	2
	B	1, 2, 3
II	A	2
	B	1, 2, 3
III	A	2, 4
	B	2, 3, 4
IV	A	2, 4
	B	2, 3, 4
V	A	2, 4
	B	2, 3, 4
VI	A	2, 4
	B	2, 3, 4
VII	A	2, 3
	B	1, 2, 3, 4

Federal Standards:

Fed. Test Method Std. No. 601—Rubber: Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information, in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

JAN–P–100 — Packaging and Packing for Overseas Shipment, General Specification.

MIL–P–116—Preservation, Methods of.

MIL–P–775 — Packaging and Packing for Overseas Shipment and Domestic Shipment Involving Storage of Hose, Rubber or Fabric (Including Tubing); and Fittings, Nozzles and Strainers.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–166—Visual Inspection Guide for Rubber Hose.

MIL–STD–417 — Rubber Compositions, Vulcanized, General Purpose, Solid (Symbols and Tests).

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

U. S. Treasury Department:

Internal Revenue Service Regulations, 26–CFR–212 — Formulas for Denatured Alcohol.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders should cite "26 CFR 212".)

3. REQUIREMENTS

3.1 Qualification. The hose furnished under this specification shall be a product which has been tested and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list (see 4.2, 6.4, and Appendix).

3.2 Materials. Hose furnished under this specification (see 6.1) shall be composed of elastomeric material(s). This material shall be used in conjunction with a fabric, yarn, or wire reinforcement as applicable (see 3.3.4).

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3.3 General construction.

3.3.1 *Types I through VI hose.* Types I through VI hose shall consist of an inner tube, an intermediate textile reinforcement, and an outer cover. When required, and as applicable (see table I), hose with one class in the tube and another in the cover may be specified (class 2 tube and class 3 cover as an example). When class 4 is specified, both the tube and cover are normally of the same class. When different classes for the tube and cover are specified, the physical properties specified for the respective parts shall apply. Adhesion requirements shall be based on the class for which the lower value is specified.

3.3.2 *Type VII hose.* Type VII hose shall consist of an inner tube, a helical, wire reinforcement, an outer cover, and shall have a preformed cuff (soft end) on each end of the hose.

3.3.3 *Inner tube.* The inner tube shall be a seamless, continuous extrusion of elastomeric material extending the full length of the hose. Minimum thickness of inner tubes of types I through VI hose shall be 0.062 inch and of type VII hose shall be 0.085 inch.

3.3.4 *Reinforcement.*

3.3.4.1 *Types I through VI hose.* The reinforcement of types I through VI hose shall consist of multiple plies of woven or cord fabric, or ply or plies of braided or knitted yarn. The reinforcement in grade A hose shall be funginert.

3.3.4.1.1 *Woven or cord fabric.* The layer or layers of fabric shall be well frictioned with elastomeric material which shall firmly join the plies to the tube and cover, and to each other.

3.3.4.1.2 *Braided or knitted yarn.* The ply or plies of reinforcement shall be braided or knitted over the tube and the cover stock shall adhere firmly to the tube stock. In multiple-ply construction there shall be a distinct layer of elastomeric material between the plies of reinforcement.

3.3.4.2 *Type VII hose.* Type VII hose shall be reinforced with coiled, high-carbon, spring-steel wire which shall not be exposed on either the inner or outer diameter of the hose.

3.3.5 *Cover.* Covers shall be bonded or vulcanized to the tubes. Covers for types I through VI hose shall have 0.032-inch minimum thickness, and minimum thickness for type VII hose shall be sufficient to adequately cover the reinforcing wire without any exposure of wire through cover.

3.3.6 *Cuffs (for type VII hose).* The preformed cuffs for type VII hose shall be built up with plies of elastomeric material or elastomeric material-impregnated fabric and shall be of uniform thickness.

3.4 Dimensions.

3.4.1 *Sizes.* Hose shall be furnished in the nominal inside diameters for which minimum burst values are shown in table II (see 6.2).

3.4.2 *Diameter tolerances.* Tolerances on hose diameters shall be as shown in tables III, IV, and V.

3.4.3 *Wall thickness, types I through VI.* Hose-wall thickness for types I through V shall be within the limits shown in table VI, and wall thickness for type VI hose shall conform to table IV.

3.4.4 *Lengths.*

3.4.4.1 *Tolerances on lengths.* Tolerances on types I through VI hose for specified (cut) lengths and on type VII hose shall be as shown in table VII.

3.4.4.2 *Cuff length (type VII).* The cuff length on each end of type VII hose shall be $1\frac{1}{2} \pm \frac{1}{8}$ inches.

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TABLE II.—*Applicability and minimum burst values of hose by types and sizes*

Type I		Type II		Type III		Type IV		Type V		Type VI		Type VII	
Nom. ID (in.)	Min. burst press. (p.s.i.)	Nom. ID (in.)	Min. burst press. (p.s.i.)	Nom. ID (in.)	Min. burst press. (p.s.i.)	Nom. ID (in.)	Min. burst press. (p.s.i.)	Nom. ID (in.)	Min. burst press. (p.s.i.)	Nom. ID (in.)	Min. burst press. (p.s.i.)	Nom. ID (in.)	Min. burst press. (p.s.i.)
3/8	475	1 1/4	500	3/8	475	1 1/4	500	3/8	90	3/8	200	1 1/4	225
1/2	425	1 1/2	450	1/2	425	1 1/2	450	1/2	90	1/2	200	1 1/2	200
5/8	360	1 3/4	400	5/8	360	1 3/4	400	5/8	90	5/8	140	1 3/4	175
3/4	325	2	350	3/4	325	2	350	3/4	90	3/4	140	2	150
7/8	300	2 1/4	350	1	275	2 1/4	350	7/8	90	7/8	130	2 1/4	125
1	275	2 1/2	300	1 1/8	250	2 1/2	300	1	90	1	125	2 1/2	100
1 1/8	250	2 3/4	250	1 1/4	225	2 3/4	250	1 1/8	90	---	---	3	75
1 1/4	225	3	250	1 5/16	215	3	250	1 1/4	90	---	---	---	---
1 5/16	215	---	---	1 1/2	200	---	---	1 1/2	90	---	---	---	---
1 3/8	1	---	---	1 9/16	190	---	---	1 3/4	80	---	---	---	---
1 7/16	1	---	---	1 3/4	175	---	---	2	70	---	---	---	---
1 1/2	200	---	---	1 7/8	165	---	---	2 1/4	60	---	---	---	---
1 9/16	190	---	---	2	150	---	---	---	---	---	---	---	---
1 5/8	185	---	---	2 1/4	125	---	---	---	---	---	---	---	---
1 3/4	175	---	---	2 1/2	100	---	---	---	---	---	---	---	---
1 7/8	165	---	---	2 5/8	95	---	---	---	---	---	---	---	---
2	150	---	---	2 3/4	90	---	---	---	---	---	---	---	---
2 1/8	140	---	---	2 7/8	85	---	---	---	---	---	---	---	---
2 1/4	125	---	---	3	75	---	---	---	---	---	---	---	---
2 3/8	115	---	---	---	---	---	---	---	---	---	---	---	---
2 1/2	100	---	---	---	---	---	---	---	---	---	---	---	---
2 5/8	95	---	---	---	---	---	---	---	---	---	---	---	---
2 3/4	90	---	---	---	---	---	---	---	---	---	---	---	---
2 7/8	85	---	---	---	---	---	---	---	---	---	---	---	---
3	75	---	---	---	---	---	---	---	---	---	---	---	---
3 1/8	1	---	---	---	---	---	---	---	---	---	---	---	---
3 1/4	65	---	---	---	---	---	---	---	---	---	---	---	---
3 3/8	1	---	---	---	---	---	---	---	---	---	---	---	---
3 1/2	50	---	---	---	---	---	---	---	---	---	---	---	---
3 3/4	1	---	---	---	---	---	---	---	---	---	---	---	---
4	1	---	---	---	---	---	---	---	---	---	---	---	---
4 1/2	1	---	---	---	---	---	---	---	---	---	---	---	---

[1] Minimum burst values to be specified by the procuring activity.

TABLE III.—*Hose diameter tolerances (types I through V)*

Inside diameter		Tolerances	
Over	To and including	Straight hose	Curved hose
Inches	*Inches*	*Inch*	*Inch*
3/8	3/4	± 1/64	± 1/32
3/4	2	+ 1/64, - 1/32	± 1/32
2	3 1/2	± 1/32	± 1/32

TABLE IV.—*Hose diameter tolerance (type VI)*

Nom. inside diameter	Tolerance	Nom. outside diameter	Tolerance
Inch	*Inch*	*Inch*	*Inch*
3/8	+ 1/64, - 3/64	11/16	± 1/32
1/2	+ 1/64, - 3/64	13/16	± 1/32
5/8	+ 1/64, - 3/64	15/16	± 1/32
3/4	+ 1/64, - 3/64	1 1/16	± 1/32
7/8	+ 1/64, - 3/64	1 5/16	± 1/16
1	+ 1/64, - 3/64	1 5/16	± 1/16

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TABLE V.—*Hose diameter tolerances (type VII)*

Inside diam. (hose and cuff)	Tolerance	Outside diam. (cuff only)	Tolerance
Inches	*Inch*	*Inches*	*Inch*
1¼	+ 1/64 – 1/32	1⅝	+ 1/16 – 0
1½	+ 1/64 – 1/32	2	+ 1/16 – 0
1¾	+ 1/64 – 1/32	2¼	+ 1/16 – 0
2	+ 1/64 – 1/32	2½	+ 1/16 – 0
2¼	± 1/32	2⅞	+ 1/16 – 0
2½	± 1/32	3⅛	+ 1/16 – 0
3	± 1/32	3¾	+ 1/16 – 0

TABLE VI.—*Wall thickness limits (types I through V)*

Designation	Thickness	
	Min.	Max.
	Inch	*Inch*
Types I, III and V	11/64	7/32
Types II and IV	15/64	9/32

TABLE VII.—*Tolerances on hose lengths*

For types I through VI, straight, and for type VII

Over	To and including	Tolerance
Inches	*Inches*	*Inch*
—	12	+ 3/16 – 1/16
12	24	+ ¼ – ⅛
24	—	+ ½ – ¼

Curved (preformed)

Each end (measured from end to intersection of nearest centerline)	Tolerance
	Inch
	+ 3/16 – ⅛

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TABLE VIII.—*Physical properties requirements (types I through VI hose)*

Physical properties	Tube requirements							
	Applicable types				Classes			
	I and II	III and IV	V	VI	1	2	3	4
Original properties								
Hardness, durometer, Shore A	X	--	--	--	55 to 75	55 to 75	NA	NA
	--	X	--	--	*NA	55 to 75	55 to 75	60 to 80
	--	--	X	X	NA	60 to 80	60 to 80	60 to 80
Tensile strength, min., p.s.i.	X	--	--	--	1000	1000	NA	NA
	--	X	X	X	NA	1000	1000	1000
Elongation, min., percent	X	--	--	--	250	250	NA	NA
	--	X	--	--	NA	225	250	225
	--	--	X	X	NA	200	225	225
Heat aging at temperatures noted								
Temperature °F.	X	--	--	--	212	212	NA	NA
	--	X	X	X	NA	212	212	250
(*Changes after 70 hours*)								
Hardness, durometer, Shore A	X	--	--	--	0 to +20	0 to +20	NA	NA
	--	X	X	X	NA	0 to +20	0 to +15	0 to +15
Tensile strength, max., percent	X	--	--	--	−20	−15	NA	NA
	--	X	X	X	NA	−15	−25	−40
Elongation, max., percent	X	--	--	--	−50	−40	NA	NA
	--	X	X	X	NA	−40	−40	−40
Oil immersion (change)								
Volume, max., percent	X	--	--	--	−5 to +25	+100	NA	NA
	--	X	X	X	NA	+100	NA	NA
Tensile strength, max., percent	X	--	--	--	−20	−60	NA	NA
	--	X	X	X	NA	−60	NA	NA

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TABLE VIII.—*Physical properties requirements (types I through VI hose)*—(Cont.)

Physical properties	Tube requirements							
	Applicable types				Classes			
	I and II	III and IV	V	VI	1	2	3	4
Coolant immersion (change)								
Volume, max., percent	X	--	--	--	0 to +20	0 to +20	NA	NA
	--	X	X	X	NA	0 to +20	0 to +15	0 to +15
Hardness, durometer, Shore A	X	--	--	--	−10 to +10	−20 to +10	−10 to +20	−10 to +20
Tensile strength, max., percent	X	--	--	--	−20	−20	NA	NA
	--	X	X	X	NA	−20	−20	−20
Elongation, max., percent	X	X	X	X	−40	−40	−40	−40

* NA—Not Applicable.

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TABLE VIII.—*Physical properties requirements (types I through VI hose)*—(Cont.)

Physical properties	Cover requirements							
	Applicable types				Classes			
	I and II	III and IV	V	VI	1	2	3	4
Original properties								
Tensile strength, min., p.s.i.	X	--	--	--	1500	1500	NA	NA
	--	X	X	X	NA	1000	1000	1000
Elongation, min., percent	X	--	--	--	225	225	NA	NA
	--	X	X	X	NA	200	225	225
Heat aging at temperatures noted								
Temperature, °F.	X	--	--	--	212	212	NA	NA
	--	X	X	X	NA	212	212	250
(*Changes after 70 hours*)								
Tensile strength, max., percent	X	--	--	--	−20	−15	NA	NA
	--	X	X	X	NA	−20	−25	−40
Elongation, max., percent	X	--	--	--	−60	−50	NA	NA
	--	X	X	X	NA	−55	−60	−60
Oil immersion (change)								
Volume, max., percent	X	--	--	--	−5 to +25	+100	NA	NA
	--	X	X	X	NA	+100	NA	NA
Tensile strength, max., percent	X	--	--	--	−45	−65	NA	NA
	--	X	X	X	NA	−65	NA	NA
Adhesion, min., pounds								
Tube to ply	X	--	--	--	12	12	NA	NA
	--	X	--	--	NA	8	8	8
	--	--	X	--	NA	10	8	8
	--	--	--	X	NA	8	6	6
Ply to ply	X	--	--	--	12	12	NA	NA
	--	X	X	--	NA	8	8	8
	--	--	--	X	NA	8	6	6

NA—Not Applicable.

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TABLE IX.—*Physical properties requirements (type VII hose)*

Physical properties	Tube	Cover
Original properties		
Hardness, durometer, Shore A	55 to 75	
Tensile strength, min., p.s.i.	1000	1500
Elongation, min., percent	250	250
Heat aging at temperature noted		
Temperature, °F.	212	212
(Change after 70 hours)		
Hardness, durometer, Shore A	0 to +20	NA
Tensile strength, max., percent	-25	-25
Elongation, max., percent	-60	-60
**Oil immerson (change)*		
Volume, max., percent	+43	+74
Tensile strength, max., percent	-65	-65
Elongation, max., percent	-50	-50
Coolant immersion (change)		
Volume, max., percent	+20	NA
Hardness, durometer, Shore A	-10 to + 10	NA
Tensile strength, max., percent	-20	NA
Elongation, max., percent	-40	NA
Adhesion (minimum pounds)		
Tube to cover	25	

* Applicable to classes 1 and 2 only.

TABLE X.—*Applicable tests*

Paragraph number	Test	Qualification	Acceptance	Control
4.5.1	Burst pressure	X	X	
4.5.2	Hardness, durometer	X	X	
4.5.3	Tensile strength and elongation	X	X	
4.5.4	Resistance to heat	X		
4.5.5	Resistance to oil	X		X
4.5.6	Resistance to coolant	X		X
4.5.7	Friction (adhesion)	X	X	
4.5.8	Low temperature flexibility	X		X
4.5.9	Ozone resistance	X		
4.5.10	Fungi resistance	X		
4.5.11	Proof pressure	X	X	
4.5.12	Resistance to vibration	X		X
4.5.13	Packaging		X	
4.5.14	Packing		X	

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3.4.4.3 *Bulk hose.* Unless otherwise specified, bulk hose of types I through VI shall be furnished in lengths of 40 to 50 feet, or multiples thereof, except that on such orders, up to 10 percent may be furnished in random lengths over 15 feet and an additional 10 percent may be furnished in random lengths over 3 feet.

3.5 Physical requirements.

3.5.1 *Burst pressure.* When tested as specified in 4.5.1, hose shall not burst at any pressure below that specified in table II.

3.5.2 *Hardness, durometer.* When tested as specified in 4.5.2, the durometer hardness of the hose tube specimens shall be as specified in tables VIII and IX, as applicable.

3.5.3 *Tensile strength and elongation.* When tested as specified in 4.5.3, hose shall meet the tensile strength and elongation requirements specified in tables VIII and IX, as applicable.

3.5.4 *Resistance to heat.* When tested as specified in 4.5.4, and as applicable (see table I), types I through VI, classes 1, 2 and 3, and type VII covers and tubes shall be oven aged at 212° F. Types III through VI, as applicable (see table I), class 4 covers and tubes shall be oven aged at 250° F. Oven aging shall be for a period of 70 hours. Hose shall conform to changes in physical properties as specified in tables VIII and IX.

3.5.5 *Resistance to oil.* When tested as specified in 4.5.5, hose shall conform to the changes in physical properties specified in tables VIII and IX as applicable.

3.5.6 *Resistance to coolant.* When tested as specified in 4.5.6, hose shall conform to changes in physical properties as specified in tables VIII and IX as applicable.

3.5.7 *Friction (adhesion).*

3.5.7.1 *Types I through VI.* When tested as specified in 4.5.7, the force required to separate tube from cover and ply from ply shall be not less than that specified in table VIII.

3.5.7.2 *Type VII.* When tested as specified in 4.5.7, the force required to separate cover from tube shall be not less than that specified in table IX.

3.5.8 *Low-temperature flexibility.*

3.5.8.1 *Types I through VI.* Hose specimens shall show no evidence of cracks or breaks when tested as specified in 4.5.8.1.

3.5.8.2 *Type VII.* When tested as specified in 4.5.8.2, hose shall show no cracks or breaks.

3.5.9 *Ozone resistance.* Types I through VI, classes 2 and 4, and type VII hose covers shall be ozone resistant. After hose has been tested as specified in 4.5.9.1.1 or 4.5.9.1.2, as applicable, the cover shall show no evidence of cracking.

3.5.10 *Fungus resistance.* Grade A hose shall be fungus resistant. After being tested as specified in 4.5.10, hose shall meet the requirements of 3.5.11.

3.5.11 *Proof pressure.* When tested as specified in 4.5.11, hose shall show no leakage.

3.5.11.1 *Pulsating proof pressure.* When tested as specified in 4.5.11.1, hose shall not bulge, separate, or show evidence of leakage.

3.5.12 *Resistance to vibration (type VII).* When tested as specified in 4.5.12, type VII hose shall show no leakage or other failure.

3.6 Marking for identification.

3.6.1 *For types I through VI.* Unless otherwise specified, each length of hose under 36 inches shall be marked on the outer surface in contrasting color, by stencil or printed type, with the following information (more than one line may be used):

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Government stock number
Nominal size (I.D.)
"OZ" (for ozone-resistant cover of class 2 or 4)
Type
Grade
Class (if cover and tube are of different classes, both classes shall be indicated and separated with a hyphen with cover class preceding tube class)
Manufacturer's name or trademark
Month and year of manufacture

On lengths of hose greater than 36 inches, the marking shall be applied continuously, using one or more lines of print, if necessary, at intervals of not more than 36 inches.

3.6.2 *For type VII.* Unless otherwise specified, the outer surface of one or both cuffs of each type VII hose shall be marked in contrasting color, by stencil or printed type, with the following information:

Government stock number
Nominal size (I.D.)
"OZ" (to denote ozone resistance)
Grade
Manufacturer's name or trademark
Month and year of manufacture

3.7 Date of manufacture. Hose shall be not more than four quarters old when submitted for acceptance.

3.8 Workmanship. The workmanship shall be such as to produce hose free from looseness of cover, exposed reinforcements (wire, cords, or fabric), cuts, bruises, blisters, breaks, pitting, wrinkles in cover, loose foreign matter and excess mandrel lubricant, and any other defects which affect hose serviceability and appearance.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Contractor inspection. The supplier is responsible for all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. The Government is responsible for examination, testing, and approval of products to be listed on the Qualified Products List.

4.2 Qualification.

4.2.1 *Types I through VI.* Hose submitted for qualification (see 3.1, 6.4, and Appendix) shall be examined for the defects listed in 4.3.2.2.2 and then the hose and platen press sheets shall be subjected to the applicable tests specified for qualification in table X.

4.2.2 *Type VII hose.* Hose submitted for qualification (see 3.1, 6.4, and Appendix) shall be examined for the defects listed in 4.3.2.2.2 and then the hose and platen press sheets shall be subjected to the applicable tests specified for qualification in table X.

4.2.3 *Failure.* Failure of a qualification test specimen to pass any specified test may be cause for refusal by the Government to conduct additional tests until it has been proved to the satisfaction of the Government activity conducting the test that the faults revealed by such test have been corrected.

4.3 Acceptance inspection.

4.3.1 *Sampling for hose.*

4.3.1.1 *Nonmilitary.*

4.3.1.1.1 *Lot formation.* Unless otherwise specified in the invitation for bid, the lot shall be formed as described in Federal Test Method Standard No. 601.

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4.3.1.1.2 *Sampling for acceptance examination.* Unless otherwise specified in the invitation for bids, the sample for examination inspection shall be taken as described in Federal Test Method Standard No. 601.

4.3.1.1.3 *Sampling for acceptance testing.* Unless otherwise specified in the invitation for bid, the sample for tests shall be taken as described in Federal Test Method Standard No. 601. Sampling plan B shall be used except that for lot sizes of 50 to 1000 units the number of test units under the first sample shall be 3 and the rejection number 2; under the second sample the number of units shall be 3; and under the combined samples the number of test units shall be 6, acceptance number 1 and rejection number 2.

4.3.1.2 *Military.*

4.3.1.2.1 *Lot formation.* Unless otherwise specified, a lot of hose shall consist of 1 type, grade, size, and class of rubber or rubber combination, as applicable (see 3.3), conforming to the requirements of this specification, from an identifiable production period from 1 manufacturer, submitted at 1 time for acceptance.

4.3.1.2.2 *Sampling for acceptance examination.* Unless otherwise specified, the Government inspector shall select a representative sample from each lot in accordance with inspection level II of Military Standard MIL–STD–105, for examination inspection purposes.

4.3.1.2.3 *Sampling for acceptance testing.* Samples for acceptance testing shall be selected in accordance with inspection level L4 of Military Standard MIL–STD–105.

4.3.2 *Examination inspection.*

4.3.2.1 *Nonmilitary.* Each unit in the sample for examination inspection (see 4.3.1.1.2) shall be inspected for construction, marking and workmanship. Dimensional measurements shall be made as specified in Federal Test Method Standard No. 601.

4.3.2.1.1 *Length.* The length shall be determined as described in method 2411 of Federal Test Method Standard No. 601.

4.3.2.1.2 *Diameter.* The average diameter shall be determined as described in method 2221 or 2231 of Federal Test Method Standard No. 601, except that 2 sets of 2 measurements, as applicable, shall be made on each test unit. If method 2231 is used, methods 2341 and 2351 shall be used for determining the inner and outer circumference respectively.

4.3.2.2 *Military.*

4.3.2.2.1 *Acceptable quality levels.* Each hose selected in accordance with paragraph 4.3.1.2.2 shall be examined for characteristics listed in the classification of defects in 4.3.2.2.2. Military Standard MIL–STD–166 shall be used as a guide for visual examination.

Classification	*AQL*
Major	2.5
Minor	4.0

4.3.2.2.2 *Classification of defects.* For examination purposes, defects shall be classified as follows:

Categories	Defects	Method of inspection
Critical	None defined	
Major:	AQL 2.5 percent	
101	Dimensions not within specified tolerances (see 3.4.2, 3.4.3, and 3.4.4)	s.i.e.
102	Incorrect cover thickness (see 3.3.5)	s.i.e.
103	Nonuniform thickness of cuffs (see 3.3.6)	s.i.e.
104	Illegible, improper or missing marking (see 3.6)	visual
105	Foreign matter or mandrel lubricant in inner tube (see 3.8)	visual
106	Looseness of cover (see 3.8)	visual
107	Exposed reinforcements (see 3.8)	visual
108	Cuts and bruises (see 3.8)	visual

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Categories	Defects	Method of inspection
109	Breaks (see 3.8)	visual
110	Blisters (see 3.8)	visual
111	Pitting (see 3.8)	visual
Minor:	AQL 4.0 percent	
201	Wrinkles in cover (see 3.8)	visual

4.3.3 *Testing inspection.*

4.3.3.1 *Acceptance tests.* Each test unit in the sample specified in 4.3.1.1.3 and 4.3.1.2.3 shall be subjected to the applicable tests shown in table X.

4.3.3.1.1 *Failure.* Failure of the samples to pass inspection or any acceptance test shall be cause for rejection of the lot represented. A test may be repeated at the request and expense of the contractor, in which case twice the original number of specimens shall be tested. Failure of any of the retest specimens shall be considered cause for rejection of the entire lot.

4.3.4 *Sampling for packaging and packing.*

4.3.4.1 *Lot formation.* A lot shall consist of all packs prepared for shipment in accordance with one level (see 5.1), from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.3.4.2 *Sampling for acceptance examination.* Sampling for acceptance examination shall be performed in the manner specified in 4.3.1.2.2.

4.3.4.3 *Sampling for acceptance testing.* When Military Specification MIL–P–775 specifies packaging of the hose, one package and one complete pack shall be selected; when packaging is not specified, only one complete pack shall be selected from each lot.

4.3.5 *Examination inspection for packaging and packing.*

4.3.5.1 *Acceptable quality level.* Each sample selected in accordance with 4.3.4.2 shall be examined for conformance to the following acceptable quality levels (AQL) on the basis of percent defective:

Classification	AQL
Major	2.5
Minor	4.0

4.3.5.2 *Classification of defects.* For examination purposes, defects shall be classified as follows:

Categories	Defects	Hose only	Hose ass'y	Method of inspection
Critical	None defined			
Major:	*AQL (2.5 percent)*			
101	Parts improperly cleaned (see 5.1.1)	X	X	visual
102	Improper preservative on fittings (see 5.1.1)		X	visual
103	Improper wrap or fittings (see 5.1.1)		X	visual
104	Wrap improperly secured (see 5.1.1)		X	visual
105	Improper interior container (see 5.1.1)	X	X	visual
106	Improper exterior container (see 5.1.1)	X	X	visual
107	Exterior container improperly sealed (see 5.1.1)	X	X	visual
108	Hose coiled in an unsafe diameter (see 5.1.1)	X	X	visual
109	Illegible marking (see 5.1.1)	X	X	visual
110	Incorrect marking (see 5.1.1)	X	X	visual
Minor:	*AQL (4.0 percent)*			
201	Improper wrapping (see 5.1.1)	X	X	visual

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4.3.6 *Testing inspection for packages and packs.*

4.3.6.1 *Acceptance testing.* When level A or B is specified (see 6.2), the samples selected in accordance with 4.3.4.3 shall be tested as specified in 4.5.13 and 4.5.14.

4.3.6.2 *Failure.* Failure of the package or pack to pass any specified test may be cause for the Government to refuse to accept the lot until it has been proved to the satisfaction of the Government that the faults revealed by the test have been corrected.

4.4 **Control tests.**

4.4.1 *Sampling for control tests of hose with class 2 or class 4 cover.* 2-foot lengths of hose with class 2 or class 4 cover shall be selected and subjected to the applicable tests shown in table X. The first sample shall be taken at or near the start of production and thereafter as deemed necessary by the Government.

4.4.2 *Failure.* Failure of the control sample to pass any test shall result in discontinuance of acceptance by the Government until evidence satisfactory to the Government has been submitted by the contractor that corrective action has been taken to eliminate the condition that caused the failure.

4.5 **Test procedures.** Except as otherwise specified herein, tests shall be made in the ambient air temperature specified in Federal Test Method Standard No. 601.

4.5.1 *Burst pressure.* To determine conformance to 3.5.1, hose test specimen shall be subjected to the burst pressure test specified in method 10011 of Federal Test Method Standard No. 601. Length of the type VII hose specimen shall be 24 inches.

4.5.2 *Hardness, durometer.* To determine conformance to 3.5.2, a test specimen from the tube of the hose being tested shall be subjected to the durometer hardness test specified in method 3021 of Federal Test Method Standard No. 601.

4.5.3 *Tensile strength and elongation.* To determine conformance to 3.5.3, tensile strength and elongation tests shall be conducted in accordance with methods 4111 and 4121 of Federal Test Method Standard No. 601. Tube and cover test specimens shall be taken from one or more samples of hose being tested. If cover sample is less than 0.031 inch in thickness, sample shall be taken from the platen press sheet for testing.

4.5.4 *Resistance to heat.* To determine conformance to 3.5.4, hose shall be subjected to the aging tests specified in either method 7221 or method 7231 of Federal Test Method Standard No. 601, at the applicable temperatures and for the periods of time shown in tables VIII and IX. Test specimens shall be taken as specified in 4.5.3.

4.5.5 *Resistance to oil.* To determine conformance to 3.5.5, specimens of classes 1 and 2 tubes and covers shall be subjected to the tensile strength portion of the immersion test specified in method 6111 and the volume change test specified in method 6211 of Federal Test Method Standard No. 601. Immersion shall be for 70 hours at 212° F. The test liquid shall be medium 3, specified in method 6001 of Federal Test Method Standard No. 601.

4.5.6 *Resistance to coolant.* To determine conformance to 3.5.6, tube specimen shall be subjected to the immersion tests specified in methods 6111 and 6211 of Federal Test Method Standard No. 601. Immersion period during the tests shall be 70 hours and the liquid shall be maintained at the boiling point under a water-cooled, reflux condenser. The immersion liquid shall be as follows:

⅓ by volume, distilled water.
⅓ by volume, alcohol (special denatured, grade No. 1) conforming to the requirements of the Appendix to Regulations No. 3, Formulae for Completely and Specially Denatured Alcohol,

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issued by the U. S. Treasury Department, Bureau of Internal Revenue.

⅓ by volume, ethylene glycol (refined grade) conforming to the following:

Acidity	Not more than 0.01 percent acetic acid.
Specific gravity	1.1150 to 1.1158 @ 20°C.
Color	Water white.
Boiling range (760-MM. pressure):	
above 210°C.	None.
below 190°C.	None.
below 195°C.	5 percent, maximum distillation.
below 202°C.	95 percent, minimum distillation.
Water	Not more than 0.5 percent by weight.
Ash	Not more than 0.005 gram per 100 cubic centimeters.
Odor	Mild, nonresidual.
Average weight	9.28 pounds per gallon at 20°C.

4.5.7 *Friction (adhesion).* Hose specimen shall be subjected to the friction test specified in method 8011 of Federal Test Method Standard No. 601 to determine conformance to 3.5.7.

4.5.8 *Low-temperature flexibility.*

4.5.8.1 *Types I through VI.* To determine conformance to 3.5.8.1, a one-inch section of hose of full circumference shall be placed in a cold box and held for 5 hours at minus 65° ± 2° F. for grade A hose and at minus 40° ± 2° F. for grade B hose. The cooling medium shall be air. While still at the specified temperature, the specimen shall be compressed between 2 parallel plates to 50 percent of its original inside diameter. The testing fixture shall be in the cold box during the entire test.

4.5.8.2 *Type VII.* To determine conformance to 3.5.8.2, one complete hose shall be conditioned for 5 hours at a temperature of minus 65° ± 2° F. for the grade A hose and at minus 40° ± 2° F. for the grade B hose. While still at this temperature, the specimen shall be bent over a curved surface having a radius equal to one and one-half times the outer diameter of the hose. Bending shall be at the rate of 15 to 18 degrees per second. Specimen shall then be straightened and bent in the opposite direction over the same curved surface at the same rate of speed. Time consumed in bending shall be held, as far as possible, to a maximum of thirty seconds.

4.5.9 *Ozone resistance.*

4.5.9.1 *Preparation of specimens.*

4.5.9.1.1 *Types I through VI hose.* Types I through VI hose with class 2 or class 4 cover only shall be tested. Cover should be removed from tube and buffed to 0.075 ± 0.005 inch thickness if necessary, or to smooth surface on thin covers. From buffed cover stock cut out duplicate specimens with 2-inch T-50 die. Insert T-50 specimens (two together) in a 2¼ inch wide, slotted, aluminum holder designed to obtain 12½ percent specimen elongation.

4.5.9.1.2 *Type VII hose.* A specimen ¾-inch wide and 6 inches long shall be cut from the platen sheet of hose cover (see 4.2.2) and attached to a specimen holder such as shown in method 7311 of Federal Test Method Standard No. 601. Specimen shall then be elongated 12½ percent.

4.5.9.2 *Procedure.* The test specimens, while still elongated, shall be conditioned for 45 minutes in air at room temperature and then placed in a test chamber maintaining air mixed with ozone in the proportion of 50 ± 5 parts of ozone per 100,000,000 parts of atmospheric air by volume. The specimen shall be exposed to the mixture of ozone and air for 168 hours, at a temperature of 100° ± 2° F. At the end of the 168-hour period, the specimen shall be examined under 9-power magnification to determine conformance to 3.5.9.

4.5.10 *Fungi resistance.* To determine conformance to 3.5.10, grade A hose shall be tested in accordance with method 5760.1 of

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Federal Specification CCC–T–191, using an incubation period of 90 days, and shall then be subjected to the test specified in 4.5.11.

4.5.11 *Proof pressure.* To determine conformance to 3.5.11, hose specimen shall be subjected to the test for proof pressure specified in method 10211 of Federal Test Method Standard No. 601. Proof pressure shall be ½ of the burst value for similar size hose shown in table II.

4.5.11.1 *Pulsating proof pressure.* To determine conformance to 3.5.11.1, the hose shall be tested as specified in method 10211 of Federal Test Method Standard No. 601. Lengths shall be as specified on applicable drawings, and the pressure shall be 40 p.s.i. applied internally at pulsating intervals of 5 seconds (2 seconds "on" and 3 seconds "off") for 15 minutes.

4.5.12 *Resistance to vibration.*

4.5.12.1 *Test specimen.* Type VII hose only shall be tested. Each specimen shall consist of a 24-inch long hose which has been conditioned for 24 hours at 80° ± 9° F.

4.5.12.2 *Procedure.* The specimen shall be installed on the vibration equipment in a manner incorporating a 90-degree bend and a 45-degree circumferential twist in the hose specimen. Vibration of the specimen shall be for 50 hours at 30 ± 10 cycles per second (c.p.s.) at ¼ inch double amplitude, and in an ambient air temperature of 80° ± 9° F. During the vibration period, water shall be circulated through the specimen at a pressure equal to ¼ of the burst value for similar size hose shown in table II to determine conformance to 3.5.12.

4.5.13 *Packaging test.* To determine conformance to method of packaging specified in Military Specification MIL–P–775, the package shall be subjected to the test specified in Military Specification MIL–P–116 for the method of packaging used.

4.5.14 *Packing test.* To determine conformance to the method of packing specified in Military Specification MIL–P–775, the level A packs shall be subjected to the interior packing test of Military Specification JAN–P–100. Level B packs shall be subjected to the same test, except the height of the drop shall be 75 percent of the height of the drop specified therein.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. Packaging shall be level A or C and packing shall be levels A, B or C as specified (see 6.2).

5.1.1 *Packaging, packing and marking.* Cleaning, packaging, packing and marking shall conform to Military Specification MIL–P–775. When tested as specified in 4.5.13 the package shall meet the requirements specified in Military Specification MIL–P–116 for the method of packaging used. When tested as specified in 4.5.14, level A packs shall meet the requirements of the interior packing test of Military Specification JAN–P–100. Level B packs shall meet the requirements of the same test, except the height of the drop shall be 75 percent of the height of the drop specified therein.

6. NOTES

6.1 Intended use. The hose covered by this specification is intended for the following uses:

(*a*) Types I and II hose are primarily intended for severe service such as on heavy-duty, military and commercial trucks, military combat vehicles, and off-the-road, earth-moving equipment.

(*b*) Type III hose is intended for normal service and passenger cars.

(*c*) Type IV hose is intended for moderate service conditions encountered with light trucks.

(*d*) Type V hose is intended for normal

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service and is of the construction most satisfactory for the manufacture of preformed, curved hose.

(*e*) Type VI hose is intended for connecting hot water heaters to the coolant-circulating systems of ground vehicles.

(*f*) Type VII hose is intended for use in engine-coolant systems in passenger cars, trucks, military-combat vehicles and off-the-road, earth-moving equipment.

6.1.1 *Other uses.* The applicable provisions of this specification may be used in the procurement of coolant system hose required for installation on industrial or marine engines.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Title, number, and date of applicable drawings (see 3.2).

(*c*) Title, number, and date of applicable military standard (see 6.8).

(*d*) Type and grade of hose, and class of rubber required (see 1.2, 1.2.1, 1.2.2, 6.3, 6.7 and 6.8).

(*e*) Size and length required (see 3.4.1, 3.4.4 and table II).

(*f*) Level of preservation and packaging, and level of packing required (see 5.1 and 5.2).

(*g*) When control tests are necessary (see 4.4.1).

6.3 Classes. The classes, as specified in 1.2.1, are described generally as follows:

Class 1 — Compounds having low volume swell in petroleum hydrocarbons. These compositions are normally made from copolymers containing acrylonitrile.

Class 2—Compounds having medium volume swell in petroleum hydrocarbons and having ozone-resistant properties. These compounds are normally made from polychloroprene as the base material.

Class 3 — Compounds that are nonoil-resistant. These compounds are normally made from natural, reclaimed or synthetic rubber, or mixtures thereof.

Class 4—Compounds having high-temperature resistant and ozone-resistant properties. These compounds normally are made from a copolymer of isoprene and isobutylene.

6.3.1 *Classes.* Comparable classes described in Military Standard MIL–STD–417 (Ord) are as follows:

ZZ–H–428b	*MIL–STD–417(Ord)*
Class 1	Class SB (Suffixes E_3 and E_5)
Class 2	Class SC (Suffixes C_1 and E_3)
Class 3	Class RN/RS
Class 4	Class RS (Suffixes A_1 and C_2)

6.4 Qualification. With respect to products requiring qualification (see 3.1, 4.2, and Appendix), awards will be made only for such products as have, prior to the bid opening date, been tested and approved for inclusion in the applicable qualified products list whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for qualified products list is Headquarters, U. S. Army Ordnance Tank-Automotive Command, 1501 Beard Street, Detroit 9, Michigan.

6.5 Transportation descriptions and minimum weights. Transportation descriptions and minimum weights applicable to this commodity are:

(*a*) **Rail:**

Hose, rubber, with or without wire re-

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inforcements.

Carload minimum weights 20,000 and 30,000 pounds, subject to Rule 34, Uniform Freight Classification.

(*b*) Motor:

Hose, rubber, reinforced with wire or fabric.

Truckload minimum weight (W) 30.6 pounds, (W) subject to Rule 34, National Motor Freight Classification.

6.6 Classification cross reference. The following is a cross reference of the classifications of ZZ–H–428 and ZZ–H–428b (the types, grades, and classes in this revision are not changed from ZZ–H–00428a(Ord)):

ZZ–H–428, Feb. 27, 1951	This specification, ZZ–H–428b
Type I, grade A, class 1	Type II, grade B, classes 1 and 2
Type I, grade A, class 2	Type I, grade B, classes 1 and 2
Type I, grade B, class 2	Type III, grade B, classes 2, 3 and 4
Type I, grade B, class 1	Type IV, grade B, classes 2, 3 and 4
Type II, grade B, class 2	Type V, grade B, classes 2, 3 and 4
Type II, grade A, class 2	No counterpart
No counterpart	Type VI, grade B, classes 2, 3 and 4
No counterpart	Type VII, grades A and B

6.7 Military hose. Types I through VI, grade A, classes 2 and 4 hose, in any applicable combination of type and class (see tables I and II), and type VII, grade A hose (see MS51008), are intended to be used for military service.

6.8 Items procured under this specification for military use are to be limited to the variety shown on the applicable military standards. Personnel of the Military Departments are requested to refer to these documents for guidance.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY CUSTODIANS:

Army—Ord
Navy—Docks
Air Force—MOA.

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APPENDIX

10. SCOPE. This appendix covers by types, the requirements for submittal of hose for qualification.

20. Submittal plans.

20.1 Types I through VI. For each type and grade of straight hose and class of rubber the manufacturer wishes to qualify (see 3.1, 4.2, 6.4, 30, and table XI), he shall submit 18 feet of hose and six platen press sheets (three each of tube and cover). Each sheet shall be 6 by 6 by 0.75 ± .005 inches. Platen press sheets shall be certified as having come from the same batch as the tube and cover of the hose submitted and as having a curing time equivalent to that of the finished products. When curved hose is being submitted for qualification, pieces of curved hose equal to 18 feet shall be furnished. Hose submitted shall be representative of that intended to be furnished under contract by the manufacturer.

20.2 Type VII hose. For each grade and size of type VII hose the manufacturer wishes to qualify (see 3.1, 4.2, 6.4, 30, and table XI), he shall submit a sample consisting of 8 complete 24-inch lengths of hose, and six platen press sheets as specified in 20.1.

30. Special provisions for establishing qualification. Qualification of types, grades, classes, and sizes of hose tabulated under "Hose qualified" in table XI, submitted by one manufacturer, will be recognized as establishing qualification of the applicable types, grades, classes and sizes of hose of the same manufacturer, tabulated under "Qualification established for" in table XI.

TABLE XI.—*Establishing qualification*

Hose qualified				Quantity for test	Qualification established for			
Type	Grade	Classes	Size*		Types	Grades	Classes	Size
II	A	2	1½	18 feet	I and II	A and B	2	All sizes
II	B	1, 3	1½	18 feet	I and II	B	1, 3	All sizes
IV	A	2, 4	1½	18 feet	III and IV	A and B	2, 4	All sizes
IV	B	3	1½	18 feet	III and IV	B	3	All sizes
V	A	2, 4	1½	18 feet	V	A and B	2, 4	All sizes
V	B	3	1½	18 feet	V	B	3	All sizes
VI	A	2, 4	⅝	18 feet	VI	A and B	2, 4	All sizes
VI	B	3	⅝	18 feet	VI	B	3	All sizes
VII	A	2, 3	2½	8, 24-inch pcs.	VII	A and B	2, 3	All sizes
VII	B	1, 4	2½	8, 24-inch pcs.	VII	B	1, 4	All sizes

* Submittal of any other size shall be subject to the approval of the qualifying activity.

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